

RAG and LangChain Deep Dive

RAG Overview

A Retrieval-Augmented Generation pipeline consists of document ingestion, text extraction, chunking, embedding generation, vector storage, similarity search, prompt construction, and answer generation.

Chunking

Recursive character splitting is commonly used. Chunk size and overlap significantly affect retrieval quality. Overlap preserves context between chunks.

Embeddings

Embedding models convert text into dense numerical vectors. Similar meanings produce vectors that are close in high-dimensional space.

Vector Databases

Popular vector databases include Pinecone, Chroma, FAISS, Weaviate, Milvus, and Qdrant. They support nearest-neighbor search using cosine similarity or Euclidean distance.

LangChain Workflow

Typical workflow: Load PDF → Split Documents → Create Embeddings → Store in Vector Database → Retrieve Relevant Chunks → Pass Context to LLM → Generate Final Answer.

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